Lightweight Machine Learning-Based IDS for IoT Environments

Zakaria Alomari, Zhida Li, Adetokunbo Makanju
Department of Computer Science, New York Institute of Technology
Vancouver, British Columbia, Canada
Email: {zalomari, zli74, amakanju}@nyit.edu

2024 8th Cyber Security in Networking Conference (CSNet) | 979-8-3315-3410-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/CSNET64211.2024.10851737

***Abstract*—This paper presents a lightweight, machine learning-based Intrusion Detection System (IDS) optimized for Internet of Things (IoT) environments, specifically designed to handle encrypted traffic. Utilizing advanced machine learning techniques, our IDS efficiently detects and analyzes malicious activities within encrypted data streams without decryption. Experimental results show that our system not only exceeds traditional IDS solutions in detection accuracy but also operates effectively under IoT constraints. Its capability to dynamically adapt to new threats and learn from continuous traffic ensures its suitability for the dynamic landscape of network security.**

***Index Terms*—Intrusion Detection System, IoT security, machine learning, encrypted traffic, network security, anomaly detection, adaptive learning, lightweight IDS.**

I. Introduction

As the Internet of Things (IoT) increasingly integrates into our daily lives and industrial processes, it heralds a new era of interconnectedness marked by enhanced efficiency and convenience. However, this expansion also introduces significant cybersecurity vulnerabilities due to the inherent limitations of IoT devices, such as limited computational resources and energy efficiency. Moreover, the growing reliance on encrypted communications, crucial for maintaining privacy, presents substantial challenges for traditional cybersecurity frameworks [1]–[3].

The evolution of Intrusion Detection Systems (IDS) within the IoT ecosystem necessitates a paradigm shift to align threat detection precision with IoT devices' operational constraints. Traditional IDS often struggle with the complexities of encrypted traffic, a challenge not fully addressed by current methodologies [4]–[7].

This paper explores the potential of a novel machine learning-based IDS tailored for the IoT ecosystem. Our approach integrates operational efficiency with comprehensive threat detection capabilities, focusing on encrypted traffic. We plan to rigorously test and validate our system across various IoT scenarios to ensure its effectiveness against the evolving threat landscape and address current IDS capabilities' critical gaps.

Our research contrasts our model with existing approaches, underscoring the enhanced detection capabilities and resource efficiencies our solution introduces, particularly in managing encrypted traffic. This analysis highlights the advancements our research contributes to IoT security and addresses significant gaps in previous studies [10]–[12].

In conclusion, our study introduces an advanced, adaptable, and efficient solution to the increasing cybersecurity challenges in the IoT ecosystem. This new standard for IoT security incorporates robust and innovative defenses tailored to the unique demands of this environment. Through this work, we aim to establish a new benchmark in IoT security, protecting interconnected devices against a broad spectrum of cyber threats and safeguarding the IoT landscape's integrity and privacy.

The remainder of this paper is structured as follows: Section II reviews related work in IDS development for IoT, focusing on the integration of machine learning for improved security measures. Section III outlines the problem statement regarding encrypted traffic handling within IoT. Section IV describes our simulation setup. Section V introduces our machine learning-based IDS solution for encrypted traffic. Section VI presents our simulation results, and Section VII concludes with future work discussions to optimize the machine learning model and enhance system performance.

This work was supported by the New York Tech-Vancouver Research Grant.

II. Related Work

This section reviews significant advancements in the development of Intrusion Detection Systems (IDS) for the Internet of Things (IoT), particularly highlighting the role of machine learning (ML) techniques in enhancing security against evolving cyber threats, especially in managing encrypted traffic.

Fenanir et al. [1] utilized lightweight ML techniques to develop IDS solutions tailored for IoT, using datasets like KDD Cup 99, NSL-KDD, and UNSW-NB15. Their research highlighted ML's effectiveness in reducing computational overhead but noted limitations in addressing new cyber threats and practical deployment.

Zhao et al. [2] advanced IDS by implementing a lightweight neural network, aiming for computational efficiency within IoT constraints. However, the model's reliance on specific datasets and limited real-world testing highlighted challenges in its adaptability and practical implementation.

Islam et al. [3] explored using deep learning networks to enhance IDS capabilities within IoT, addressing computational and storage limitations. Despite advancements, their work revealed gaps in handling encrypted traffic, suggesting a need for more adaptable analytical frameworks.

Adnan et al. [12] critically evaluated ML-based IDS, pointing out issues with data storage and computational efficiency. Their insights emphasized the need for IDS that manage high-dimensional data without depending on traditional data models.

Jan et al. [10] simplified intrusion detection by focusing on single data attributes, like packet arrival rates, proving that minimalistic approaches could be effective. This method highlighted the necessity for broader validation mechanisms to ensure IDS effectiveness across various IoT configurations.

Building on these studies, our research proposes a machine learning-based IDS tailored for IoT, incorporating advanced feature selection techniques to address encrypted traffic. Our model aims to overcome previous limitations in dataset relevance, algorithmic flexibility, and deployment practicality. Through extensive testing, we demonstrate our framework's enhanced ability to manage encrypted traffic, setting a new standard in IDS performance for IoT security.

III. Problem Statement

The expansion of the Internet of Things (IoT) brings both increased connectivity and significant cybersecurity challenges. As IoT devices grow in number, their vulnerabilities, particularly with encrypted traffic, also rise. While encryption enhances privacy, it complicates threat detection.

This paper introduces a lightweight Intrusion Detection System (IDS) that uses machine learning to improve threat detection in encrypted traffic without compromising privacy. Current efforts to enhance IDS efficiency face limitations, such as reliance on specific datasets and challenges in real-world deployment, affecting scalability and adaptability [1], [10]–[12].

Our novel IDS model applies machine learning to encrypted IoT traffic, improving analysis and adapting to changing network conditions, effectively detecting hidden threats. This approach combines machine learning's detection power with encryption, ensuring strong security while maintaining privacy, even in resource-constrained IoT devices.

By integrating machine learning with encryption, this method provides a proactive defense for securing IoT data, offering both security and efficiency in encrypted traffic analysis. Traditional IDSs often struggle with encrypted data, delaying threat detection and lowering classification accuracy [4], [5]. These limitations highlight the need for innovative solutions like ours.

Our approach enhances IDS capabilities by combining the strengths of machine learning and encryption, safeguarding IoT transmissions while maintaining data confidentiality. This dual strategy improves the detection of complex threats in encrypted traffic, surpassing traditional methods.

The IoT ecosystem urgently needs advanced security solutions for encrypted traffic. Our machine learning-based IDS addresses these needs, offering enhanced privacy, network protection, and intrusion detection efficiency. This solution sets a new standard in defending against evolving threats.

IV. Simulation

This study evaluates the performance of a Machine Learning-based Intrusion Detection System (IDS) on encrypted and unencrypted IoT traffic. By comparing both types of traffic, the experiment aims to assess how machine learning and encryption improve intrusion detection accuracy and efficiency.

A. Experimental Setup

The experimental setup comprises two virtual machines, representing an IoT source device and an Intrusion Detection System (IDS). The source device virtual machine employs iperf to generate network traffic, including both encrypted and unencrypted data streams, simulating the data transmission from IoT devices. The IDS virtual machine is equipped with Snort for intrusion detection and traffic analysis, alongside the machine learning model for traffic classification and potential security threat identification.

B. Procedure

The experiment involves four stages: unencrypted and encrypted traffic are generated using iperf and analyzed by Snort on the IDS virtual machine, with threats, false positives, and accuracy recorded. Machine learning is then integrated into the analysis of both unencrypted and encrypted traffic, comparing the results with the initial analyses to assess improvements in detection accuracy.

C. Data Collection and Analysis

At each stage, collect metrics such as *detection accuracy*, *false positive rate*, *processing latency*, and *CPU/memory usage* to assess performance. Compare the results to evaluate the impact of encryption and machine learning on IDS performance.

By analyzing different scenarios, we aim to show how integrating machine learning and encryption improves IDS, particularly for encrypted IoT traffic, providing insights for more efficient and accurate detection systems.

V. Solution

This paper presents a lightweight machine learning-based Intrusion Detection System (IDS) designed for managing encrypted traffic in IoT environments. Through extensive testing on real network traffic, we compare the performance of traditional IDS with machine learning-enhanced IDS in detecting threats in both encrypted and unencrypted scenarios.

Our findings demonstrate that machine learning significantly improves IDS detection capabilities, particularly for encrypted traffic, although the degree of enhancement depends on the choice of models, feature extraction techniques, and training data quality. We also note a potential trade-off with operational efficiency, highlighting the need for careful optimization for practical deployment.

This study underscores the potential of machine learning to enhance intrusion detection for encrypted traffic and identifies challenges that must be addressed to improve IDS precision and efficiency. Future research will focus on refining machine learning algorithms and feature engineering to better detect and mitigate threats within encrypted traffic.

A. Performance Comparison between Traditional IDS and Machine Learning-enhanced IDS

1) Normal Traffic Analysis:
Traditional IDS effectively detect known threats using signature-based methods but struggle with new or modified attacks. Machine learning-enhanced IDS, however, adapt better by learning from updated datasets. Both systems operate at similar speeds, showing that machine learning doesn't reduce efficiency. Though training machine learning models is resource-intensive, their operational costs remain comparable to traditional IDS, enhancing threat detection without raising expenses.

2) Encrypted Traffic Analysis:
Traditional IDS face challenges with encrypted traffic, as encryption hides packet payloads they rely on inspecting. This reduces their effectiveness in environments with prevalent encryption, highlighting the need for IDS that adapt to modern encryption without compromising threat detection.

3) Machine-Learning Filtering of Normal Traffic:
Integrating machine learning into IDS frameworks improves detection accuracy and reduces false positives and negatives. While processing time slightly increases, the approach lowers error rates and adapts to new threats, enhancing network security.

4) Machine-Learning Filtering of Encrypted Traffic:
Machine learning-enhanced IDS require more computational resources for encrypted traffic analysis but excel at detecting advanced cyber threats. While this increases CPU and memory usage, it enhances protection where encryption hides malicious activities. Despite these challenges, machine learning represents a key advancement in cybersecurity, emphasizing the need for ongoing model refinement to balance accuracy and efficiency.

B. Selection and Training of the Machine Learning Model

To enhance Intrusion Detection Systems (IDS) for encrypted traffic, we selected the RandomForestClassifier for its strong performance in complex data scenarios and effective feature importance evaluation. This model excels in environments with non-linear and obscured data relationships, typical of encrypted traffic. By employing an ensemble of decision trees, it reduces overfitting and improves accuracy while handling large datasets with diverse features such as packet sizes, timing intervals, and protocol types.

The flexibility of the RandomForestClassifier allows for easy scalability and adaptation to new threats without complete retraining, which is crucial in dynamic IDS environments. Its capability to assess feature importance is vital for identifying key indicators of anomalies in encrypted traffic.

By averaging predictions across multiple trees, the model prevents overfitting and maintains effectiveness on unseen data, thereby enhancing detection accuracy and operational efficiency—making it suitable for modern cybersecurity challenges.

Our training process focused on key features, including protocol types, packet sizes, and transmission times. Extensive data preprocessing ensured dataset integrity and optimized conditions for machine learning. The RandomForestClassifier achieved an impressive accuracy rate of 92.8%, significantly improving its ability to detect malicious activities in encrypted traffic.

Table I demonstrates the model's high Precision, Recall, and F1-Scores across various classes, showcasing its proficiency in accurately classifying benign and malicious traffic in encrypted data streams. The high F1-Scores reflect a balanced measure of Precision and Recall, underscoring the classifier's consistency in real-world IDS scenarios.

TABLE I: Model Accuracy Rate.

	Precision	Recall	F1-Score	Support
0	0.90	0.95	0.92	7418
1	0.96	0.91	0.93	9049
Accuracy			0.93	16467
Macro Avg	0.93	0.93	0.93	16467
Weighted Avg	0.93	0.93	0.93	16467

Integrating this machine learning-enhanced IDS has produced promising results in real-time traffic analysis, providing a dynamic approach to threat detection. Moving forward, we plan to continuously refine our model to adapt to evolving network behaviors and encryption methods, ensuring its effectiveness against emerging security threats and maintaining robust cybersecurity measures.

C. Lightweight IDS for Encrypted IoT Traffic

This paper introduces a Lightweight Intrusion Detection System (LIDS) for IoT environments, using machine learning to detect encrypted traffic anomalies. Encryption protocols like SSL/TLS obscure payloads, making detection difficult. LIDS addresses this by analyzing traffic metadata with advanced algorithms, avoiding the need for payload inspection.

Optimized for resource-limited IoT devices, LIDS detects abnormal patterns in encrypted data, signaling potential threats by learning typical behaviors and spotting deviations. It also streamlines data processing for efficient and accurate threat detection in environments with limited resources.

Fig. 1 shows the "LIDS Encrypted Traffic Detection Workflow," where encrypted traffic is transmitted to the IDS, which uses machine learning to analyze and detect attacks. Alerts are generated for threats, while normal traffic is passed through. This illustrates LIDS's effectiveness in securing IoT networks using real-time machine learning analysis.

D. Implementation of Real-Time Traffic Monitoring

Our study developed a real-time traffic monitoring system using Scapy for network packet capture and machine learning models for traffic analysis. Scapy handles continuous traffic streams, while the machine learning model, trained on normal and malicious patterns, accurately distinguishes between benign and harmful activities.

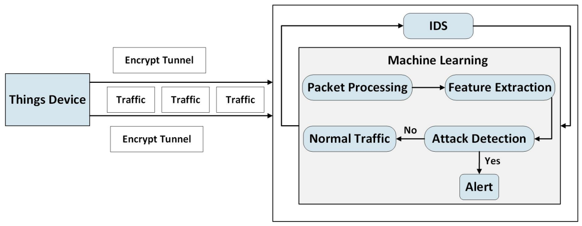

Fig. 1: Lightweight IDS for Encrypted IoT Traffic.

Key challenges included maintaining performance under high traffic volumes and ensuring threat detection for emerging attacks. We optimized the machine learning model and continuously updated the training datasets to adapt to evolving threats and encryption methods.

Deploying the system in real-world environments required network tuning and performance optimization. We implemented a robust framework to ensure real-time analysis without performance degradation, effectively handling encrypted traffic.

The following pseudocode illustrates a continuous monitoring loop that captures and analyzes packets in real-time. Each packet is examined at the IP layer, with essential features, including metadata for encrypted traffic analysis, extracted. The *MachineLearningModel.predict()* function then classifies the traffic as benign or malicious, triggering alerts for any detected threats to enable prompt preventive actions.

Additionally, the pseudocode demonstrates the system's advanced techniques for monitoring network traffic. The *extract_features()* function efficiently captures key data, including encryption patterns and metadata, rather than relying solely on payload content. This approach enhances the machine learning model's ability to classify traffic and strengthens defenses against cyberattacks using encrypted communications.

```
# Pseudocode for Real-Time Traffic Monitoring System
While True:
   packet = Scapy.capture_packet()
   if 'IP' in packet.layers:
      features = extract_features (packet)
      predication = MachineLearningModel.predict (features)
      if prediction == 'malicious':
         raise_alert (packet)
      else:
         log_activity (packet)
   else:
      log_error ("Non-IP_packet_detected")
```

VI. Result

In this section, we evaluate the proposed solution through extensive simulations, comparing its detection accuracy, response time, and resource utilization against a baseline. The results show improved threat detection and mitigation in encrypted traffic, highlighting the advantages of integrating machine learning into IDS for enhanced data privacy and rapid response.

A. Experimental Environment

Our research utilized a Linux virtual machine, Scapy for packet manipulation, and Scikit-learn for machine learning models. We designed experiments to evaluate an IDS with machine learning on both encrypted and unencrypted traffic, comparing its performance to traditional IDS methods in terms of efficiency and accuracy in detecting threats in encrypted data.

B. Comparative Analysis

This paper highlights the advancements in integrating machine learning with traditional IDS, particularly regarding encrypted traffic. While traditional IDS relies on signature-based detection, which is effective for unencrypted data, it struggles significantly with encrypted traffic due to its inability to inspect packet payloads.

Enhanced Detection Capabilities in Encrypted Traffic: Our machine learning-enhanced Intrusion Detection System (IDS) significantly improves threat detection in encrypted traffic. Unlike traditional systems that cannot interpret encrypted payloads, our model analyzes traffic metadata and indirect indicators of malicious activity using machine learning algorithms. This allows for the identification of threats even when content is encrypted, providing a key advantage in confidentiality-sensitive environments. Fig. 2 compares the accuracy of traditional and machine learning-enhanced IDS for both encrypted and unencrypted traffic, showing that traditional systems perform adequately in unencrypted scenarios but struggle with encrypted data, highlighting the superior capabilities of our approach.

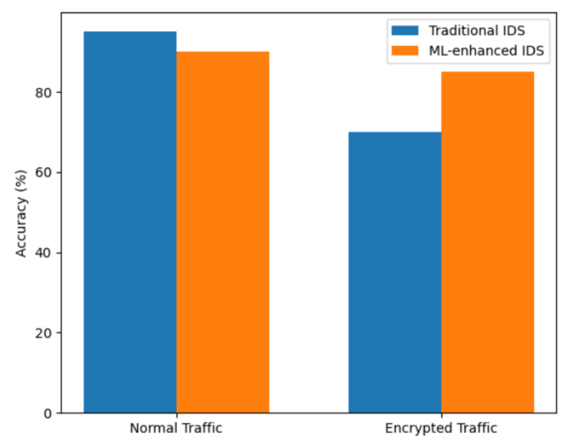

Fig. 2: Accuracy by traffic type and IDS type.

Machine Learning Integration: Our IDS uses a pre-trained machine learning model to quickly classify network traffic as normal or malicious, leveraging patterns from extensive data analysis. Unlike traditional IDS, which rely on fixed rules or signatures, this approach better detects new or advanced threats. Fig. 3 compares the workflows of traditional IDS with our machine learning-based system, highlighting the enhanced decision-making enabled by real-time analysis and adaptive algorithms.

While our machine learning-enhanced IDS shows slightly higher CPU and memory usage compared to traditional IDS as shown in Fig. 4, it offers unmatched flexibility. Despite traditional IDS systems having lower resource demands, our system's dynamic learning capabilities make it more responsive to emerging threats. Ongoing optimizations will reduce

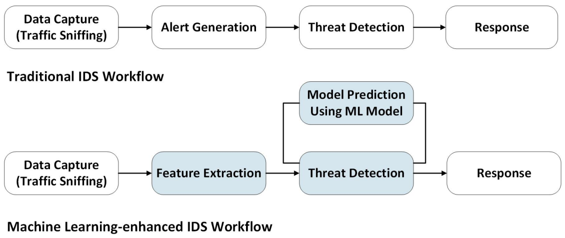

Fig. 3: Comparative Workflow Analysis.

resource consumption, making it more suitable for constrained environments.

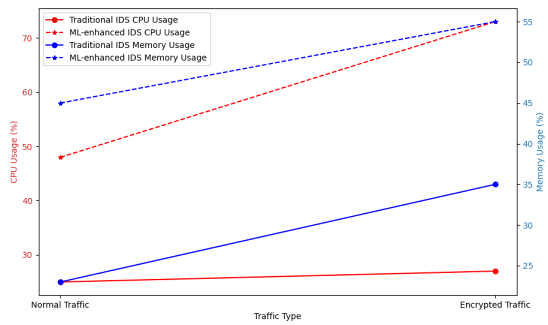

Fig. 4: Performance Comparison.

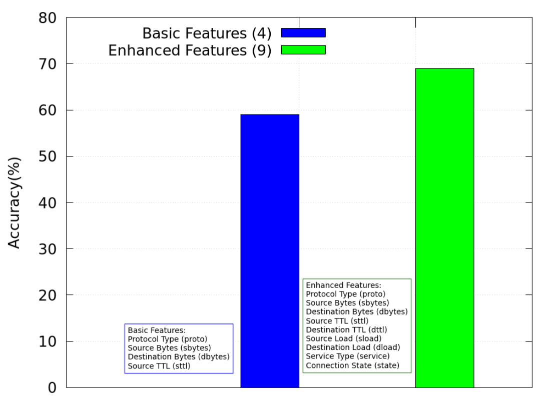

Fig. 5: Improvement in IDS Accuracy Through Feature Set Expansion and Machine Learning Integration.

Feature Enhancement and Performance Impact:

Initially, our Intrusion Detection System (IDS) utilized a basic feature set focused on protocol type *(proto)*, byte sizes *(sbytes and dbytes)*, and Time-To-Live *(sttl)* values, resulting in a real-time monitoring accuracy of only 59%. To enhance our detection capabilities, particularly for encrypted traffic, we expanded our feature set. Key additions include identifying the application-layer protocol *(service)* to differentiate traffic types, assessing connection states *(state)* to detect suspicious behaviors, measuring connection duration *(dur)* to identify potential scanning or data exfiltration, evaluating differential TTL *(dttl)* to reveal evasion tactics, and analyzing load per duration *(sload and dload)* to detect unusual data flows, such as data exfiltration or DDoS attacks.

As shown in Fig. 5, the comparison of the basic and expanded feature sets underscores the significant impact of feature expansion, which improved our system's real-time monitoring accuracy from 59% to 69%. The new features enabled our Intrusion Detection System (IDS) to more effectively distinguish between benign and malicious activities, thereby reducing false positives and false negatives.

Our approach shows that incorporating a wider range of descriptive and behavioral indicators significantly improves the effectiveness of Intrusion Detection Systems (IDS) in detecting sophisticated cyber threats, enhancing sensitivity to anomalies within encrypted data streams.

VII. Conclusion

This paper highlights the capabilities of our machine learning-enhanced Intrusion Detection System (IDS) in managing encrypted traffic. By employing advanced techniques, the system can identify threats within encrypted channels, offering significant implications for cybersecurity. This research lays the groundwork for developing more resilient and efficient security systems that can adapt to emerging challenges.

References

[1] S. Fenanir, F. Semchedine, and A. Baadache,"A Machine Learning-Based Lightweight Intrusion Detection System for the Internet of Things," Revue d'Intelligence Artificielle, vol. 33, no. 3, 2019.

[2] R. Zhao, G. Gui, Z. Xue, J. Yin, T. Ohtsuki, B. Adebisi, and H. Gacanin, "A Novel Intrusion Detection Method Based on Lightweight Neural Network for Internet of Things," IEEE Internet of Things Journal, vol. 9, no. 12, pp. 9960-9972, 2022.

[3] N. Islam, F. Farhin, I. Sultana, M. S. Kaiser, M. S. Rahman, M. Mahmud, and G. H. Cho, "Towards Machine Learning Based Intrusion Detection in IoT Networks," Computers, Materials and Continua, vol. 69, no. 2, 2021.

[4] X.-H. Nguyen, X.-D. Nguyen, H.-H. Huynh, and K.-H. Le, "Realguard: A Lightweight Network Intrusion Detection System for IoT Gateways," Sensors, vol. 22, 432, 2022.

[5] V. Pham, E. Seo, and T. M. Chung, "Lightweight Convolutional Neural Network Based Intrusion Detection System," J. Commun., vol. 15, no. 11, pp. 808-817, 2020.

[6] S. Roy, J. Li, B.-J. Choi, and Y. Bai, "A Lightweight Supervised Intrusion Detection Mechanism for IOT Networks," Future Generation Computer Systems, vol. 127, pp. 276–285, 2022.

[7] King J, Awad AI (2016) A distributed security mechanism for resource-constrained IoT devices. Informatica (Slovenia) 40(1):133–143.

[8] Shaker Alanazi, Jalal Al-Muhtadi, Abdelouahid Derhab, Kashif Saleem, Afnan N. AlRomi, Hanan S. Alholaibah, Joel J.P.C. Rodrigues, On resilience of Wireless Mesh routing protocol against DoS attacks in IoT-based ambient assisted living applications, in: 2015 17th International Conference on E-Health Networking, Application Services, HealthCom, IEEE, 2015, pp. 205–210.

[9] D. Evans, "The Internet of Things - How the Next Evolution of the Internet is Changing Everything," CISCO white Pap.,no. April, pp. 1–11, 2011.

[10] Jan, S. U., Ahmed, S., Shakhov, V.Koo, I. (2019). Toward a lightweight intrusion detection system for the Internet of things. IEEE Access, 7, 42450–42471.

[11] Kayode Saheed, Y., Idris Abiodun, A., Misra, S., Kristiansen Holone, M.Colomo-Palacios, R. (2022). A machine learning-based intrusion detection for detecting Internet of things network attacks. Alexandria Engineering Journal, 61(12), 9395–9409.

[12] Adnan, A., Muhammed, A., Abd Ghani, A. A., Abdullah, A.Hakim, F. (2021, June 4). An intrusion detection system for the Internet of things based on Machine Learning: Review and Challenges. MDPI.